Development Roadmap Document

Project: Web-Based Educational Platform for Centralized Academic Resources **Document type:** Development Roadmap **Version:** 1.0

1. Overview

This roadmap sets out how the platform will be built, from planning to launch and beyond. It is organized into phases with milestones, deliverables, responsibilities, risks and success measures. Timelines assume a small student team working part-time; adjust the week counts to your academic calendar.

2. Roadmap at a Glance

Phase	Name	Duration	Key outcome
0	Planning and Setup	Weeks 1–2	Requirements, structure and tools finalized
1	Design	Weeks 3–4	UI designs, database schema, architecture approved
2	Core Development (MVP)	Weeks 5–10	Working student portal and admin dashboard
3	Testing and Hardening	Weeks 11–12	Bugs fixed, security and usability verified
4	Deployment and Launch	Week 13	Live site with initial content
5	Enhancement Release	Weeks 14–19	Search, filters, dashboard stats, backups
6	Growth and Future Features	Beyond week 19	Video, forums, AI assistant, student accounts

3. Phase 0: Planning and Setup (Weeks 1–2)

Goals: Confirm scope, collect the academic structure, and prepare the working environment.

Task	Output
Finalize requirements with guide or faculty	Approved Project Requirements Document
Collect syllabus: semesters, subjects, modules	Structured list for seeding the database
Gather sample content (notes, papers, lab manuals) and check usage rights	Initial content set
Choose technology stack	Stack decision record
Set up Git repository, branching rules and task board	Repository and board ready
Assign roles	Responsibility matrix

Milestone M0: Scope and plan approved.

4. Phase 1: Design (Weeks 3–4)

Goals: Decide how it looks and how it is built before coding.

Task	Output
User flows for student and admin	Flow diagrams
Wireframes for all main pages (desktop and mobile)	Wireframe set
Visual style: colors, fonts, components	Simple style guide
Database schema design	ER diagram and schema script
Architecture and route plan	Approved System Architecture Document
Security and upload rules	Written checklist

Milestone M1: Design sign-off.

5. Phase 2: Core Development, MVP (Weeks 5–10)

Development is run in three two-week sprints.

Sprint 1 (Weeks 5–6): Foundation and Admin Basics

- Project scaffold, environment configuration, database and migrations
- Admin authentication (login, logout, session handling)
- Admin management of semesters, subjects and modules
- Base layout, navigation, header and footer

Sprint goal: An admin can log in and build the academic structure.

Sprint 2 (Weeks 7–8): Content Upload and Student Browsing

- Resource upload with validation, storage and metadata
- Resource edit, replace and delete
- Student pages: home, semester, subject, module and resource lists
- PDF viewer and download for all supported formats
- Lab resources and question paper sections

Sprint goal: An admin can upload; a student can browse, view and download.

Sprint 3 (Weeks 9–10): Communication and Polish

- Announcements (admin and student sides)
- Feedback and contact forms with admin inbox
- Responsive design and accessibility fixes
- Error pages, loading and empty states
- Seed real content and write the first user guide

Sprint goal: Feature-complete MVP.

Milestone M2: MVP feature complete (all P1 features from the Feature Breakdown).

6. Phase 3: Testing and Hardening (Weeks 11–12)

Activity	Details
Functional testing	Test cases for every P1 feature, covering student and admin flows
Security testing	SQL injection, XSS, CSRF, unauthorized access to admin routes, malicious file uploads
Usability testing	5–10 students try tasks (find a question paper, download a lab manual) and give feedback
Compatibility testing	Chrome, Firefox, Edge, Safari; Android and iOS devices
Performance check	Page load times, large file downloads, pagination
Bug fixing	Resolve all critical and high issues; document known minor ones
Documentation	Admin manual, README and deployment notes

Milestone M3: Release candidate approved with no open critical bugs.

7. Phase 4: Deployment and Launch (Week 13)

Task	Output
Provision hosting, domain and HTTPS	Production environment
Configure production database and file storage	Secure configuration
Deploy application and run smoke tests	Live site
Upload verified initial content for at least one or two semesters	Populated platform
Set up backups and basic monitoring	Operational safeguards
Announce to students and gather early feedback	Launch communication

Milestone M4: Public launch.

8. Phase 5: Enhancement Release (Weeks 14–19)

Based on early feedback, build the P2 features.

Feature	Weeks
Basic keyword search	14–15
Filters and sorting by type, year and format	15–16
Publish/unpublish toggle, multi-file upload	16–17
Admin dashboard statistics	17
Automated backups and password change	18
About, FAQ, privacy and copyright pages	18
Regression testing and release	19

Milestone M5: Version 1.1 released.

9. Phase 6: Growth and Future Features (Beyond Week 19)

These items are subject to future development, as stated in the project abstract. Suggested order, based on value and effort:

Order	Initiative	Effort	Value	Prerequisites
1	Student accounts and bookmarks	Medium	Medium	Authentication extension
2	Advanced search with full-text indexing	Medium	High	PDF text extraction, search engine
3	Video lectures	Medium	High	Video hosting or object storage
4	Discussion forums with moderation	High	Medium	Student accounts
5	AI-powered academic assistant	High	High	Indexed content, LLM integration, usage and cost controls

Order	Initiative	Effort	Value	Prerequisites
6	PWA or mobile app	Medium	Medium	Stable API

10. Milestone Summary

Milestone	Target week	Exit criteria
M0 Plan approved	2	Scope, stack and team roles agreed
M1 Design sign-off	4	Wireframes, schema and architecture approved
M2 MVP complete	10	All P1 features working in staging
M3 Release candidate	12	Test pass, no critical bugs
M4 Launch	13	Site live with real content
M5 Version 1.1	19	P2 features released

11. Team Roles (Suggested)

Role	Responsibilities
Project lead	Planning, coordination, communication with guide and stakeholders
Front-end developer	UI implementation, responsiveness, accessibility
Back-end developer	APIs, business logic, authentication, file handling
Database and DevOps	Schema, migrations, deployment, backups
QA and documentation	Test plans, bug tracking, user guide, report
Content coordinator	Collect, verify and organize resources

Roles can be combined in smaller teams.

12. Development Practices

- Git with feature branches and pull request reviews
- Weekly progress meetings and a short review at the end of each sprint
- Task board with columns: Backlog, In Progress, Review, Testing, Done
- Continuous integration to run tests on every push
- Regular staging deployments for early feedback
- Definition of Done as listed in the Feature Breakdown Document

13. Risk Management

Risk	Likelihood	Impact	Mitigation
Scope creep	High	High	Freeze MVP scope at M1; park new ideas in the Phase 5–6 backlog
Delay in content collection	Medium	High	Start content gathering in Phase 0; accept a partial launch (one or two semesters)
Copyright concerns	Medium	High	Verify permissions, add takedown contact, prefer faculty-approved material
Team availability during exams	High	Medium	Plan buffer weeks; keep tasks small and well-documented
Security flaws in uploads	Medium	High	Early implementation of validation and dedicated security testing in Phase 3
Hosting or storage limits	Medium	Medium	Size caps per file; plan migration path to object storage
Low student adoption	Medium	Medium	Promote through class groups, collect feedback, prioritize most-requested content

14. Success Metrics

Metric	Target after 3 months
Published resources	At least 150 across covered semesters
Monthly active visitors	50% of target student population
Average time to find a resource	Under 1 minute in usability tests
Student satisfaction (feedback rating)	4 out of 5 or higher
Admin upload time per file	Under 2 minutes
Uptime	99% or better
Critical bugs after launch	Zero open beyond one week

Adjust the targets to the actual size of your institution and department.

15. Post-Launch Maintenance

- Weekly content review and uploads, with removal of outdated material
- Monthly review of feedback and prioritization of requested improvements
- Quarterly dependency and security updates
- Semester-start checklist: add new subjects, archive old announcements, verify backups
- Annual review of the roadmap and technology choices

16. Next Steps

1. Confirm the technology stack and team roles.
2. Collect the syllabus structure for semesters, subjects and modules.
3. Set up the repository and task board.
4. Create wireframes for the home page, module page and admin upload form.
5. Begin Sprint 1 once design sign-off (M1) is complete.